

Vaughn Khouri

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Profile

First-year student studying Mechanical Engineering & Robotics at the Massachusetts Institute of Technology. I'm a detail-oriented maker who loves solving difficult, multidisciplinary problems. I've led teams through every stage of the design process, from setting precise design goals to prototyping, designing, and deploying hardware in the field. I bring experience working on ME and Systems for everything from autonomous machines to robots and electric vehicles.

Experience

Brakes Subsystem, Dyno lead (*MIT Motorsports FSAE Team*)

Since 09/2025

- Leading architecture, design & fabrication of a custom dynamometer platform
 - Measures torque, allowing characterization of our motors and inverters under load
- Size, design, simulate, and validate custom-machined brake rotors and mounting buttons in Siemens NX. Fabricate rotors & buttons in-house.
- Present several stages of design reviews to Drivetrain Team members & alumni

Founding Engineer, Platform Development (*Acre Robotics*)

Summer 2025

- Led development for the MVP testing platform at Acre Robotics, a Bay Area-based Series A startup building low-cost autonomous tractors. Involved retrofitting a full size Case 75A tractor with custom hardware
- Designed and installed hardware (Actuators, cameras, GPS, housings, etc.), electronics (NVIDIA Jetson & ESP32) to enable autonomous capabilities
- Completed a full-size, field-test ready MVP in less than 30 days on a 4-figure budget

Team Captain (*FIRST Robotics Competition Team 8033*)

09/2021 - 07/2025

- Led growth from a brand new team to a robust organization with 50+ students from 17+ high schools, placing in the top 16 worldwide. Top 15 in CA 2022-2025.
- Set design requirements, lead trainings & all-team meetings, prototype, design, & iterate subsystems, and drove robot at four world championships
- Honored as a Dean's List Finalist

Apple ETC I/II Student (*Apple*)

Summer 2023, 2024

- One of fifteen selected for two on-site summer programs at Apple
- Practiced hardware design, EE, firmware development, app development, and other topics in PD through sessions with Apple mentors
- Led teams to develop fully functional hardware and software prototypes of unique product concepts, and pitched them to Apple executives
 - Responsible for hardware, firmware & electronics for "Flex," a portable cable workout machine
 - Responsible for hardware on "PowerPlay," an interactive playground concept
- Mentored Apple Engineering Camp I students

Events Planning Lead (*MIT Entrepreneurship Club*)

Since 09/2025

- Manage relationships with VCs and visiting speakers
- Plan and promote events & workshops for students that draw 200+ attendees

Education

Massachusetts Institute of Technology

08/2025 - 06/2029

B.S. Mechanical Engineering

- Specialized in Control, Instrumentation & Robotics (Course 2A-CIR)
- Planned Minor in Materials Science and Engineering (Course 3)
- GPA: 5.0/5.0

Piedmont High School

09/2021 - 07/2025

- Weighted GPA: 4.37/4.00

Skills

More info on my website, www.vkhour.dev

CAD / CAM Tools

PTC Onshape, Siemens NX (including FEA), Solidworks, Fusion 360

Machine Tools

Manual/CNC Mill, Lathe, MIG & TIG Welding, Waterjet / Fiber Laser, FDM / SLA / SLS 3D printing, All common wood / metalworking tools

Programming Languages

Python / MicroPython, Java, ROS2

Other

Conversational Mandarin (including reading / writing), First Aid & Outdoorsmanship (Eagle Scout), PCB Design & Microsoldering

Further Commitment

Encouraging Engineering (<https://www.encouragingengineering.org>)

Since 2023

- Founded an organization that writes, tests and distributes lesson content for low-cost, scalable engineering workshops, reducing average cost and overhead by 2-3x
- International organizations use our materials to run hands-on events with students
- Invited to present at the 78th UN General Assembly Science Summit
 - “Sustainable models for STEM Education with a focus on marginalized groups”

FRCDesign (<https://www.frcdesign.org>)

Since 2024

- Cofounded the first extensive online textbook for FIRST Robotics hardware design
- Contributed extensively to authoring core curriculum
- Drove substantial growth from launch to 7000+ active users across 21+ countries